11 The height h cm of a sunflower plant t days after planting the seed is modelled by $h = a + b \ln t$ for $t \geqslant 9$, where a and b are constants. The sunflower is 10 cm tall 10 days after planting and 200 cm tall 85 days after planting.

(a) (i) Show that the value of b which best models these values is 88.8 correct to **3** significant figures. **[2]**

(ii) Find the corresponding value of a. **[1]**

(b) (i) Explain why the model is not suitable for small positive values of t. **[1]**

(ii) Explain why the model is not suitable for very large positive values of t. **[1]**

(c) Show that the model indicates that the sunflower grows to 1 m in height in less than half the time it takes to grow to 2 m. **[2]**

(d) Find the value of t for which the rate of growth is 3 cm per day. **[3]**